

# POCKETSMART AI

Your Smart Budget & Recommendation Assistant

**UNIVERSITY NAME**

Madras University

**DEPARTMENT**

Department of Computer Applications

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**PROGRAMME**

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**GitHub Repository:**

<https://github.com/Afreen-Fathima697/PocketSmart-Ai-Your-smart-budget-and-Recommendation-assistant->

**Demo Video Link:**

<https://drive.google.com/file/d/1Ip1KONnP2tjbWnFd073jdXyDMO3Ati-g/view?usp=drivesdk>

**ABSTRACT**

PocketSmart AI - Your Smart Budget & Recommendation Assistant is a web-based project developed to demonstrate the practical use of generative artificial intelligence for budget-conscious planning. The system allows users to enter a budget together with requirements and preferences and receive AI-assisted suggestions through a structured web interface.

The project focuses on three planning scenarios: home interior planning, party planning, and jewelry recommendations. The application combines a Python-based backend, web technologies, database storage, environment-based configuration, and Google Gemini for generative recommendations. User authentication and recommendation history are included to provide a more organized experience.

## 1. Brainstorming & Ideation Phase

### 1.1 Project Context & Motivation

Managing a limited budget while planning purchases or events can require considerable time and comparison. Users may have to search across multiple websites, compare prices, understand product or service descriptions, and decide which alternatives suit their needs.

PocketSmart AI brings these requirements into a single planning workflow. Instead of treating the budget as an isolated value, the system combines the user's budget with contextual information such as room requirements, event details, guest count, occasion, or style preferences. A generative AI service is then used to prepare contextual recommendations.

### 1.2 Naan Mudhalvan Project Context

This project is presented as a Naan Mudhalvan project-based learning activity. The report emphasizes practical application of technical skills rather than only theoretical study. The development process involves requirement identification, system planning, programming, AI/API integration, database usage, interface development, testing, documentation, and presentation.

Through PocketSmart AI, the team applies programming and software-development concepts to a real-world style problem. The project also provides practical exposure to generative AI, environment variables, backend routing, database operations, user interaction, and modular application design.

### 1.3 Problem Statement

Users planning home improvements, parties, or occasion-based purchases may need to compare many alternatives while staying within a fixed budget. Conventional searching requires repeated manual comparison and may not combine all user requirements in one place. The project therefore aims to provide a unified AI-assisted interface in which users can specify their budget and relevant requirements and obtain organized recommendations based on the information provided.

### 1.4 Key Objectives

- To develop a user-friendly web application for budget-based planning.
- To use generative AI for producing contextual and personalized recommendations.
- To provide separate planning modules for home interiors, parties, and jewelry.
- To combine budget information with user preferences and category-specific requirements.
- To implement user registration, login, sessions, and recommendation history where applicable.
- To demonstrate integration between a Python web application and a generative AI API.
- To use environment variables for configuration values and API credentials.

## 2. Requirement Analysis Phase

### 2.1 Existing vs. Proposed System

**Existing System:** In a conventional approach, users independently search shopping websites, service platforms, blogs, or search engines. They compare alternatives, estimate costs, and manually decide which choices satisfy their requirements. This process can be repetitive and may not provide a single contextual view of the user's budget, preferences, and planning requirements.

**Proposed System:** PocketSmart AI provides a unified AI-assisted planning workflow. The user selects a planner, enters the required information, and submits the request. The application processes the input and prepares the context needed by the recommendation service. Gemini is then used to generate a response, which is presented through the application interface.

### 2.2 Functional Requirements

- The system should allow a user to register and log in where authentication is enabled.
- The system should provide access to the available planning modules (Home Interior, Party, Jewelry).
- The system should collect budget and category-specific inputs.
- The system should validate required inputs before processing.
- The system should send suitable context to the configured Gemini service.
- The system should display the generated recommendation in a readable form.
- The system should store or retrieve recommendation history where implemented.

### 2.3 Non-Functional Requirements

- **Usability:** The interface should be understandable and easy to navigate.
- **Performance:** Normal requests should be processed without unnecessary delay.
- **Reliability:** Invalid inputs and service failures should be handled appropriately.
- **Security:** Sensitive API credentials should not be hard-coded in public source files.
- **Maintainability & Scalability:** Modules should be organized so that future changes and additional categories are easier.

### 2.4 Hardware & Software Requirements

| Type                  | Requirement Details                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware Requirements | Computer or laptop capable of running Python and a modern web browser; sufficient storage for project files and database; internet connection for Gemini API access.       |
| Software Requirements | Python 3.x, Visual Studio Code / IDE, Modern Web Browsers (Chrome, Edge, Firefox), SQLite support, `python-dotenv`, required dependencies, valid Gemini API configuration. |

### 3. Project Design Phase

#### 3.1 System Architecture

PocketSmart AI can be viewed as a set of connected layers. The presentation layer collects user information and displays results. The application layer manages routes, validation, authentication, sessions, and planner logic. The AI layer prepares the recommendation context and communicates with Gemini. The data layer stores relevant application information using SQLite. A configuration layer loads environment-specific values and sensitive credentials.

- **User Interface:** Collects budget, preferences, and planner-specific information.
- **Backend / Application Layer:** Validates input, manages routes, sessions, and business logic.
- **AI Service Layer:** Prepares prompts/context and communicates with Gemini.
- **Database Layer:** Stores user and recommendation-related data.
- **Configuration Layer:** Loads values such as API keys without hard-coding secrets.

#### 3.2 Database Design

SQLite is used as a lightweight database option for local development. Depending on the final implementation, the database can maintain user-related information and recommendation history:

- **User-related data:** Used for authentication and account information.
- **Recommendation-related data:** Used to maintain submitted requirements and generated results.
- **Session/application data:** Managed through the application framework or supporting storage.

#### 3.3 Modules Specification

| Module Name               | Description & Purpose                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------|
| User Authentication       | Handles user registration, login, logout, and session-related operations.                          |
| Home Interior Planner     | Collects budget, room type, style preferences, and required items to generate home plans.          |
| Party Planner             | Uses event type, budget, guest count, venue requirements, food, and decor to format event budgets. |
| Jewelry Planner           | Accepts budget, occasion, and style preferences to suggest 22K/18K gold and diamond choices.       |
| AI Recommendation Service | Formats user requirements into structured AI prompts and processes Gemini API responses.           |
| Recommendation History    | Maintains relevant previous recommendation information for user review.                            |

## 4. Project Planning Phase

### 4.1 Scope of the Project

The scope of the project is to demonstrate generative AI in practical budget and lifestyle planning scenarios. It is intended as a recommendation and decision-support application rather than a financial transaction platform.

- Home interior budget planning and item recommendations.
- Party or event budget planning based on event requirements.
- Jewelry recommendations based on budget, occasion, and style preferences.
- User account and recommendation history functionality.
- Generative AI integration for contextual responses.

### 4.2 Project Workflow

1. Open the PocketSmart AI web application.
2. Register or log in when authentication is required.
3. Select the required planner (Home, Party, or Jewelry).
4. Enter the budget and planner-specific requirements.
5. Submit the planning request.
6. The backend validates and processes the request.
7. The application prepares the AI recommendation context.
8. The request is sent to Gemini through the configured API.
9. The generated response is processed and displayed to the user.
10. Relevant information is stored for recommendation history.

### 4.3 Security & Configuration Strategy

- Store sensitive values in a `.env` file rather than directly in source code.
- Keep `GEMINI_API_KEY` and other secrets outside publicly shared source files.
- Use a secure secret key for application sessions.
- Validate user input before processing requests.
- Do not expose internal error details to end users in a production deployment.

## 5. Project Development Phase

### 5.1 Technologies & Implementation Details

- **Python:** Application and backend logic implementation.
- **Python Web Framework:** Request handling, routing, sessions, and web application functionality.
- **SQLite:** Lightweight local database storage.
- **Google Gemini API:** Generative AI recommendation service integration.
- **HTML & CSS:** Structure and presentation of the web interface.
- **JavaScript:** Client-side interaction where required.
- **Jinja / Server-Rendered Templates:** Dynamic web page rendering.
- **python-dotenv:** Loading environment variables from configuration files.

### 5.2 Gemini AI Integration Implementation

The project uses Google Gemini as the generative AI component. User inputs are converted into a contextual request containing information such as budget and planner-specific requirements. Gemini processes this context and generates a recommendation response.

The API key is loaded through an environment variable such as ``GEMINI_API_KEY`` instead of being written directly into application source code. This separation helps reduce accidental exposure of credentials during development and version control.

AI-generated recommendations are intended as decision-support information. Product prices, availability, and other live information should be verified through appropriate current sources before making real purchases or bookings.

### 5.3 Project Structure & Organization

The project follows a modular structure in which configuration, application routes, AI/recommendation logic, database-related components, templates, and static resources are separated. This organization makes the project easier to understand, debug, test, and extend.

## 6. Project Testing Phase

### 6.1 Testing Methodology & Execution

Testing was performed on each major component before final submission. The following areas were systematically checked using the application:

- **Authentication Testing:** Registration and login with valid and invalid inputs.
- **Form Validation Testing:** Required-field validation in planner forms.
- **Input Handling:** Budget and numerical input verification.
- **API Integration Testing:** Gemini API request and response handling.
- **Error & Exception Handling:** Behavior when the API key is missing or the AI service is unavailable.
- **Database Operations:** Database insertion and retrieval for user records and history.
- **UI & Formatting:** Recommendation display, table formatting, and markdown rendering.
- **Navigation:** Navigation between major pages, modules, and logout handling.

### 6.2 Limitations & Constraint Evaluation

- AI-generated suggestions may not always represent current prices or live availability.
- Recommendation quality depends on the information provided by the user and the AI service model.
- Gemini-powered functionality requires internet connectivity and valid API configuration.
- Live product or service information requires authorized and current external data sources.
- The current project is primarily an academic prototype and requires additional deployment security for production use.

## 7. Project Documentation Phase

### 7.1 Advantages of PocketSmart AI

- Provides one unified interface for multiple lifestyle planning scenarios.
- Uses generative AI to interpret contextual user requirements.
- Combines budget information with specific user preferences.
- Reduces repetitive manual comparison during initial planning.
- Uses a modular structure that can be easily extended.
- Demonstrates practical integration of AI, backend development, and database concepts.

### 7.2 Future Enhancements

- Integration with authorized live product and service APIs.
- Additional budget-planning categories (e.g., travel, education).
- Advanced budget analytics and charts.
- Improved filtering and comparison features.
- Downloadable PDF planning reports.
- Multilingual interaction support.
- Cloud deployment and production-grade authentication.

### 7.3 Learning Outcomes

The project provided practical exposure to problem identification, requirement analysis, Python programming, web application development, database operations, API integration, generative AI concepts, environment-based configuration, testing, debugging, teamwork, documentation, and project presentation.

### 7.4 Conclusion & References

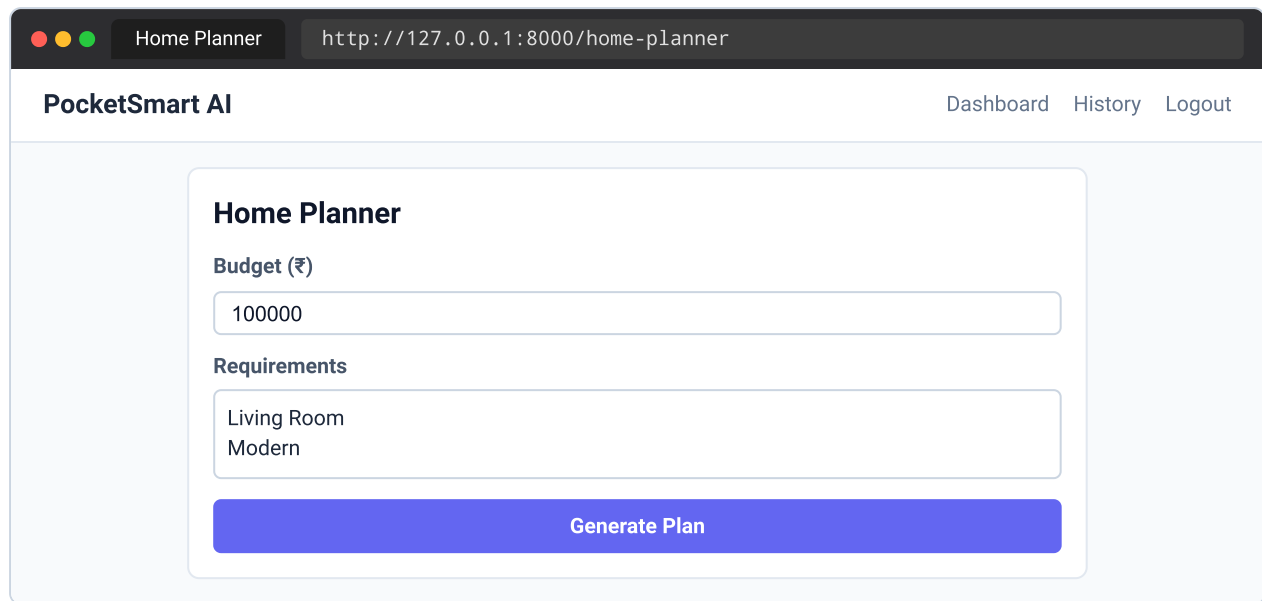
PocketSmart AI demonstrates a practical application of generative AI for budget-conscious lifestyle planning. By combining a Python-based web application, database storage, web technologies, secure configuration, and Gemini-powered recommendations, the project provides a unified workflow for several planning scenarios.

**References:** Google Gemini API / Developer Documentation; Python Documentation; SQLite Documentation; HTML, CSS, JavaScript, and Jinja Documentation; `python-dotenv` Documentation.

## 8. Project Demonstration Phase — Project Screenshots

The following real application screenshots demonstrate the working web interface and user flows as recorded from the live local server running at 127.0.0.1:8000.

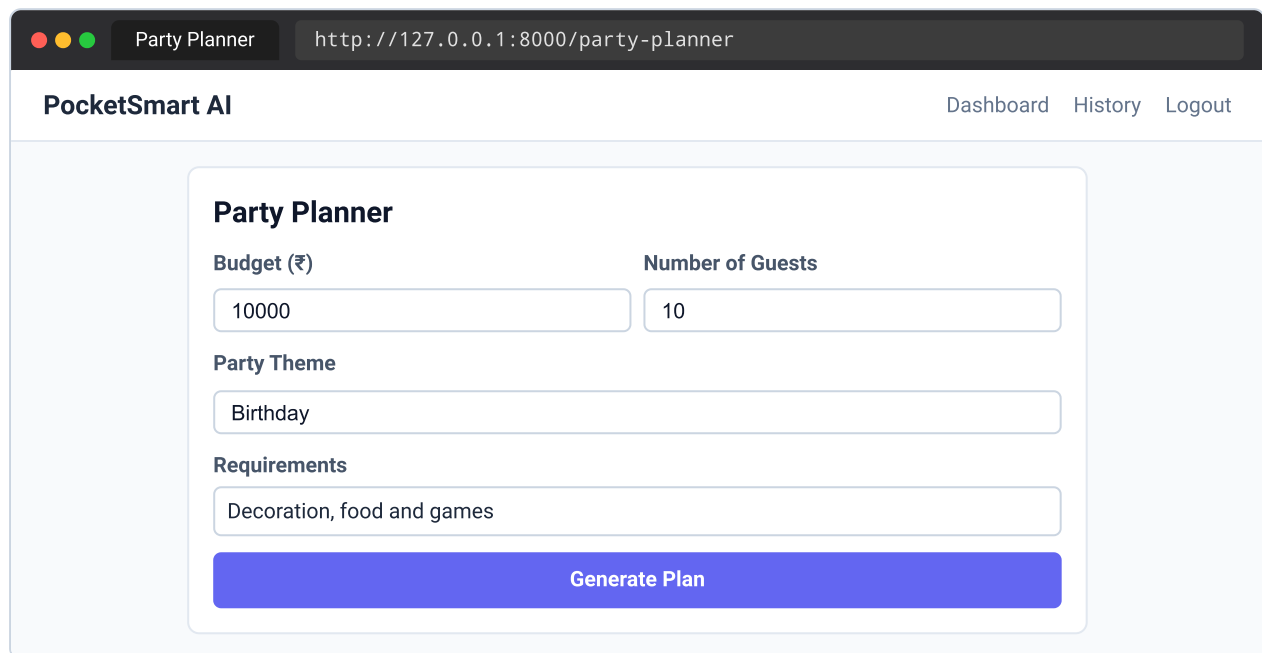
### 8.1 Home Planner Input Interface



The screenshot shows a web browser window with the title 'Home Planner' and the URL 'http://127.0.0.1:8000/home-planner'. The page header includes 'PocketSmart AI' on the left and 'Dashboard History Logout' on the right. The main content area is titled 'Home Planner' and contains the following fields:

- Budget (₹)**: A text input field containing '100000'.
- Requirements**: A text input field containing 'Living Room' and 'Modern' on separate lines.
- Generate Plan**: A large blue button.

### 8.2 Party Planner Input Interface



The screenshot shows a web browser window with the title 'Party Planner' and the URL 'http://127.0.0.1:8000/party-planner'. The page header includes 'PocketSmart AI' on the left and 'Dashboard History Logout' on the right. The main content area is titled 'Party Planner' and contains the following fields:

- Budget (₹)**: A text input field containing '10000'.
- Number of Guests**: A text input field containing '10'.
- Party Theme**: A text input field containing 'Birthday'.
- Requirements**: A text input field containing 'Decoration, food and games'.
- Generate Plan**: A large blue button.

## 8. Project Demonstration Phase (Continued)

### 8.3 AI Recommendation Result — Home Planner Output

Home Planner - Resulthttp://127.0.0.1:8000/home-planner

PocketSmart AI

DashboardHistoryLogout

Your Plan

Hello! I am "PocketSmart AI", your practical home budget and recommendation assistant. Transforming your living room into a sleek, modern space on a ₹1,00,000 budget is completely achievable. The modern aesthetic prioritizes clean lines, functional furniture, neutral palettes, and minimal clutter.

**1. Suggested Plan Design Concept:** Modern Minimalist with Warm Accents (Base neutrals paired with warm natural wood tones).

**2. Estimated Budget Allocation:**

| Category        | Description                                  | Estimated Cost (₹) |
|-----------------|----------------------------------------------|--------------------|
| Primary Seating | 3-Seater Fabric Sofa + 1 Accent Lounge Chair | 42,000             |
| Media & Storage | Low-profile TV unit / media console          | 14,000             |
| Tables          | Nesting / round coffee table set             | 8,000              |

### 8.4 Authentication Interfaces (Login & Register Screenshots)

http://127.0.0.1:8000/login

Login

Email

test@example.com

Password

.....

Login

http://127.0.0.1:8000/register

Create Account

Name

TEST USER

Email

test@example.com

Register

## 8. Project Demonstration Phase (Continued)

### 8.5 AI Recommendation Result — Party Planner Output

Party Plan Result

http://127.0.0.1:8000/party-planner

PocketSmart AI

DashboardHistoryLogout

Your Party Plan

Hello! I am "PocketSmart AI", your practical party budget and recommendation assistant. Hosting a memorable birthday party for 10 guests with a budget of ₹10,000 allows for an intimate, stylish, and high-energy celebration.

1. Suggested Party Plan:

Setting: Aesthetic living room or terrace/clubhouse setup. Duration: 3.5 to 4 hours (Evening 06:30 PM to 10:30 PM).

2. Estimated Budget Allocation:

| Category             | Description                                                | Estimated Cost (₹) |
|----------------------|------------------------------------------------------------|--------------------|
| Birthday Cake        | 1 kg fresh cream artisanal cake                            | 1,000              |
| Main Meal & Starters | Appetizers, dinner mains, and ice cream                    | 3,800              |
| Beverages            | DIY mocktail ingredients, sodas, and juices                | 900                |
| Decorations          | DIY balloon arch kit, banner, fairy lights, photo backdrop | 1,600              |
| Games & Prizes       | Game printables, props, and winner hampers                 | 1,200              |

### 8.6 Jewelry Planner Input Interface

http://127.0.0.1:8000/jewelry-planner

Jewelry Planner

Budget (₹)

100000

Requirements

Necklace

## 8. Project Demonstration Phase (Continued)

### 8.7 AI Recommendation Result — Jewelry Planner Output

Jewelry Planner - Resulthttp://127.0.0.1:8000/jewelry-planner

PocketSmart AI

DashboardHistoryLogout

Jewelry Planner — ₹1,00,000.0

Hello, I am "PocketSmart AI", your practical budget and recommendation assistant. With a budget of ₹1,00,000.00, you are in a great range to purchase either an elegant, lightweight solid gold necklace or a premium everyday-wear diamond pendant necklace.

**1. Suggested Plan & Target Category:**  
Lightweight 22K Yellow Gold Necklace (approx. 11-13 grams) OR an 18K Gold Diamond Pendant Necklace. Primary Objective: Maximize material value, aesthetic versatility, and long-term resale value.

**2. Estimated Budget Allocation:**

| Component                                 | Share (%) | Approximate Amount (₹) |
|-------------------------------------------|-----------|------------------------|
| Net Metal & Stone Cost (Gold/Diamonds)    | ~82%      | ₹82,000.00             |
| Making Charges / Craftsmanship (~12%-15%) | ~15%      | ₹15,000.00             |

### 8.8 Recommendation History View Screenshot

http://127.0.0.1:8000/history

PocketSmart AI

DashboardHistoryLogout

History Logs

Jewelry Planner — ₹1,00,000.0

Submitted requirements: Necklace | Status: Saved to SQLite database

Party Planner — ₹10,000.0

Submitted requirements: Birthday Party, 10 Guests | Status: Saved to SQLite database

## 27. PROJECT TEAM AND GUIDE

### Project Team Members

- **Student 1 (Lead):** AFREEN FATHIMA (Reg No: 212505047)
- **Student 2:** Anishapearly M
- **Student 3:** Asma K

### Academic & Program Details

- **University:** Madras University
- **Institution:** Shri Krishnaswamy College for Women, Chennai
- **Department:** Department of Computer Applications
- **Programme:** Bachelor of Computer Applications (BCA)
- **Class / Section:** II BCA 'A'
- **Skill Development Program:** Naan Mudhalvan Skill Development Programme

## 28. DECLARATION

We hereby declare that the project titled "**PocketSmart AI - Your Smart Budget & Recommendation Assistant**" has been prepared as part of the Naan Mudhalvan project-based learning/skill development activity under **Madras University**.

The information, implementation details, screenshots, and results included in this report are intended to represent the final project developed by the student/team.

**Date:** Academic Year 2026–2027

**Place:** Chennai

**AFREEN FATHIMA (Lead)**

Anishapearly M

Asma K

Signature of Students

